

Juneau: Arctic Gateway for Transitioning Towards Sustainable Development

Abstract

With the rapid development of tourism, achieving a harmonious balance among the interests of the government, tourists, and local residents has become increasingly crucial. Juneau, one of the best tourist destinations in Alaska, is caught in a dilemma of government tourism revenue and the upper limit of environment capacity. We develop a model for **sustainable tourism industry** in order to balance different levels of interests. By considering the premise of low carbon, our model optimize the government revenue from tourists, meanwhile it can also be understood to improve residents' satisfaction as much as possible on the premise of ensuring certain investment in infrastructure.

The optimized traveling model is a **multi-objective programming problem** based on the logic of a **genetic algorithm**, where the government tourism revenue and resident satisfaction are set to be maximized while the carbon emission and infrastructure maintenance costs must be minimized. According to this idea, genetic algorithm is used to solve the model.

For Problem 1, we construct an innovative **CRITIS Assist TOPSIS (CA-TOPSIS) model**, and use the **Pareto Front** derived by **NSGA-II** algorithm to obtain the optimal solution, so as to realize the redistribution of additional government income, thus reflecting the idea of sustainability. On the premise of ensuring the steady growth of government tourism revenue, we have effectively achieved the goal of energy conservation and emission reduction, and also ensured the investment in infrastructure.

For Problem 2, we built a model of **Entropy Weighted Monte Carlo Method (EMCM)** based on more random data. Problem 2 is a generalization of problem 1 model to realize the operation and management of tourist allocation in a more complex tourism environment. We used the **entropy weight method** to make a reasonable proportion of each objective function, and used the **Monte Carlo algorithm** to iterate 50 cycles, and carried out the optimization of NSGA-II in each iteration (200 iterations), thus generating $(200 \times 50 =) 10,000$ solutions, and selecting the optimal solution from these solutions to complete the **two-stage optimization** of the objective function.

The modeling process of Problem 1 and Problem 2 are innovative, where we use alternate evaluation and optimization methods. Besides, **Bayesian Thought (Prior Parameter)** was reasonably applied in the process of model operation, this improve the interpretability of the model.

From the perspective of distribution, both models have their **unique social values**. The first model concentrates on the allocation of government extra revenue, while the second model concerns reasonable scheduling of urban tourist distribution. In the social environment where **sustainability issues** are increasingly valued, the two models provide optimized reference schemes for government policy evaluation and intelligent decision-making.

keywords; Sustainable Development Tourism; Multi-objective Optimization; NSGA-II; Pareto Front; Monte Carlo Method; Evaluation Models

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1 Introduction

1.1 Background & Literature Review

In Juneau, visitors are captivated by the stunning views of the Mendenhall Glacier, where the breathtaking blue meets the lush green forest. Juneau is a city for the recreation and travel industry with substantial revenue of \$375 million in 2023. [1]. As is the case with all polar tourist destinations, the city's **capacity** has been stretched to the limit by **overexploitation** of tourism resources. In addition, due to the implications of global warming, the Mendenhall Glacier is experiencing a process of degeneration. Davies et al. studied the long-term ice loss from Juneau Icefield, indicating the areas shrinkage rates were five times faster from 2015-2019 than from 1979-1990. [2]

The government of Juneau also notices such changes. They would like to build up a **sustainable tourism industry** to balance local tourism revenue, carbon emissions, facility maintenance costs and local residents' satisfaction.

Since the Juneau locals are eager to construct a stable equity distribution model, previous research demonstrated that there are numerous models that have been used for sustainability. One specific and widely used model is the Triple Bottom Line framework created in the last century. The Triple Bottom Line expands the companies' purpose to improve its environmental performance, where they can turn the environment game into a win-win-win business strategy. [3]

Concurrently, Nash Equilibrium (NE) and Multiobjective Optimization (MOO) are mainstream approaches in case study. The two methods above have different perspectives. However, NE is sometimes unstable comparing with traditional multiobjective optimization approaches, where the solution usually is found in a Pareto Optimum. [4]

1.2 The Description of the Problem

The problem is to construct a sustainable model of tourism in Juneau. In this task, we should consider:

- Build a sustainable development model as a whole, and lock the system target as
 - Maximize Tourism Revenue (W)
 - Minimize Carbon Emission (Environment Influence) (F)
 - Maximize Social Satisfaction (S)
 - Minimize Investment of Infrastructure Maintenance (I)
- Sensitivity analysis of coefficient, influence factors and constraints of the model.
- Establish an evaluation model for extra revenue distribution.
- Generalize the sustainable model to other tourist destinations, design the model modification mechanism and define the application conditions.
- Develop a tourism resource optimization model to balance visitor distribution more intelligently.

Generally, the problem could be analyzed in five parts above.

1.3 Our work

For the first three parts above, we design a hybrid model of **NSGA-II** and **Pareto Front Model** to fit the four targets. Meanwhile, we test the model by **changing the coefficients of each formula that is optimized** and **adjusting the scope of constraints** to discover the sensitivity of the model. We use an innovative algorithm for the evaluation model: we combine the Criteria Importance Through In-criteria Correlation Weighting Method (**CA-TOPSIS** :in order to analyze importance weights between different optimized objectives) and the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS)to rank all solutions in **Pareto Front** with the updated weights between objective functions. Since it is an organic combination of two distinct algorithms, we call it **CA-TOPSIS** (Critic-Assisted TOPSIS model).

For the last two parts, we apply our model into Los Angeles, a vibrant city in Southern California known for its entertainment industry. We generalize our hybrid model into tourism industry in metropolitan with a very different urban texture comparing with Juneau. We also establish a sensitivity test and evaluation model for our generalization. Since the CA-TOPSIS is less robust in stochastic data, we established another innovative model: EMCM (Entropy-based Monte Carlo Method) model which is based on **Entropy Weighted Model** and the **Monte Carlo Algorithm** in order to find the optimized distribution of visitor numbers in different tourist attractions in LA.

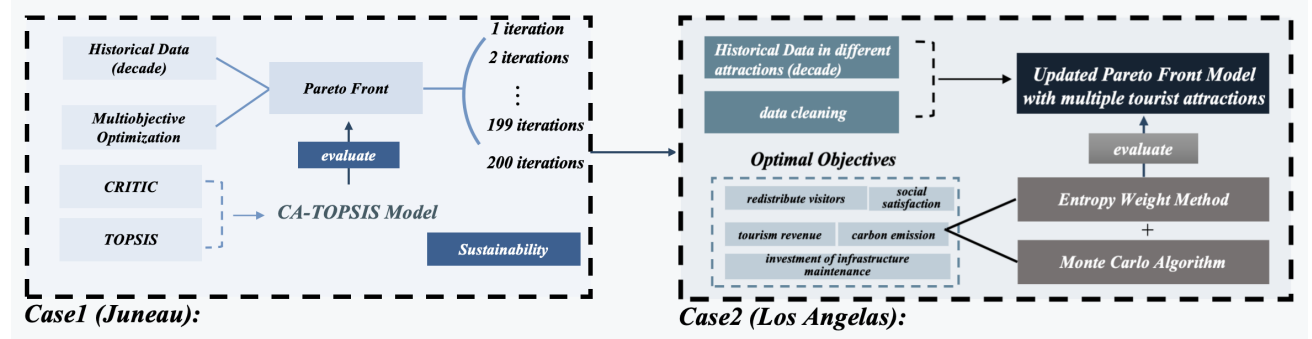


Figure 1: Flowchart of the whole modeling process

2 Assumptions and justifications

2.1 Assumption 1

We use appropriate regression model to complete the missing data between ten years for each indicator.

In general, we search for data related to the tourism industry in Juneau. We download the data on the local population, local employed population, and local population income to prepare for further analysis of local government revenue and tourism income allocation. [5] Alongside, we also find the data of number of visitors in Juneau, resident satisfaction and Carbon Dioxide Emission (in Metric Tons of CO₂ Equivalent) from 2010 to 2019, government tourism revenue and infrastructure maintenance cost from 2014 to 2023. [6–8]

Since there are two sets of data based on different time periods, we use the best fitting method to

conclude the trend of each data over time from 2010 to 2023.

According to the number of visitors in Los Angeles, the data is complete, we don't need to create fitting curve to fill the "missing data". Besides, the approximate data of carbon emission in Los Angeles is also obvious. [6] In addition, we use regression model to complete the satisfaction rate of residents from 2010 to 2015 with data 2016-2024. [9]

2.2 Assumption 2

Per capita spending of visitors in Los Angeles.

After checking the ticket prices for 6 tourist attractions we consider (three popular destinations, three unpopular attractions) in Los Angeles, average hotel spend and per capita meal spend, we can sum up the per capita spending of popular and unpopular attractions that we need to include in our algorithm.

2.3 Assumption 3

Upper bearing capacity of six attractions in Los Angeles.

After simulating the passenger flow volume for 6 tourist attractions in Los Angeles we consider, we could predict the maximum number of guests in each destinations. Overall, although it is easy to guess such data in Hollywood and Disneyland (because both of them has their tourist control model, the estimation of relatively unpopular destinations may suffer a bias.

2.4 Assumption 4

Data Cleaning

There are two mainstream data cleaning method, the 3σ Rule and the **Interquartile Range (IQR)**. Since the 3σ Rule is sometimes too loose to verify the range of outliers, which may cause more bias in our model, we choose to use the IQR rule to clean data. Two figures below are effect pictures.

Besides, Data for years affected by the pandemic were excluded.

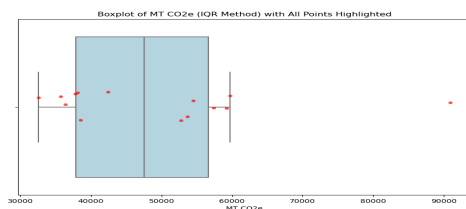


Figure 2: IQR-data cleaning in Juneau (example).

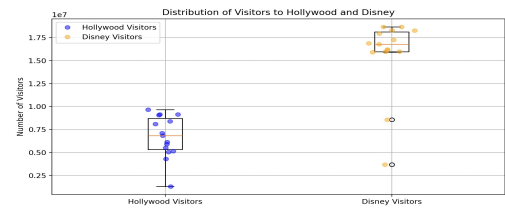


Figure 3: IQR-data cleaning in Los Angeles (example).

In 2020, the **COVID-19** epidemic swept the world, and the tourism industry of Juneau and Los Angeles were greatly affected, so the number of tourists, tourism income and even carbon emissions in the two years of 2020, 2021 are tending to be abnormal. Under such circumstance, we need to classify these data as **outliers** and ignore them from the modeling process.

2.5 Assumption 5

There is a strong negative correlation between glacier degeneration and carbon emission.

Using the long-term data of the degenerating rate of Mendenhall Glacier [2] and the carbon emission data we collect, we could calculate the relationship between these two regression model (the y-axis records Total Glacier Area per km^2 carbon emission per MT CO_2e respectively). The **Pearson's Coefficient** could quantify the predicted negative relationship.

$$r(f, g) = \frac{n \sum f(x)g(x) - \sum f(x) \sum g(x)}{\sqrt{[n \sum f(x)^2 - (\sum f(x))^2][n \sum g(x)^2 - (\sum g(x))^2]}}$$

In our situation, we define Year as x , Total Glacier Area as $f(x)$, and CO_2 emission as $g(x)$. With the coefficient of **-0.9744**, we have more than 99% confidence to say there is a negative relationship between these two data.

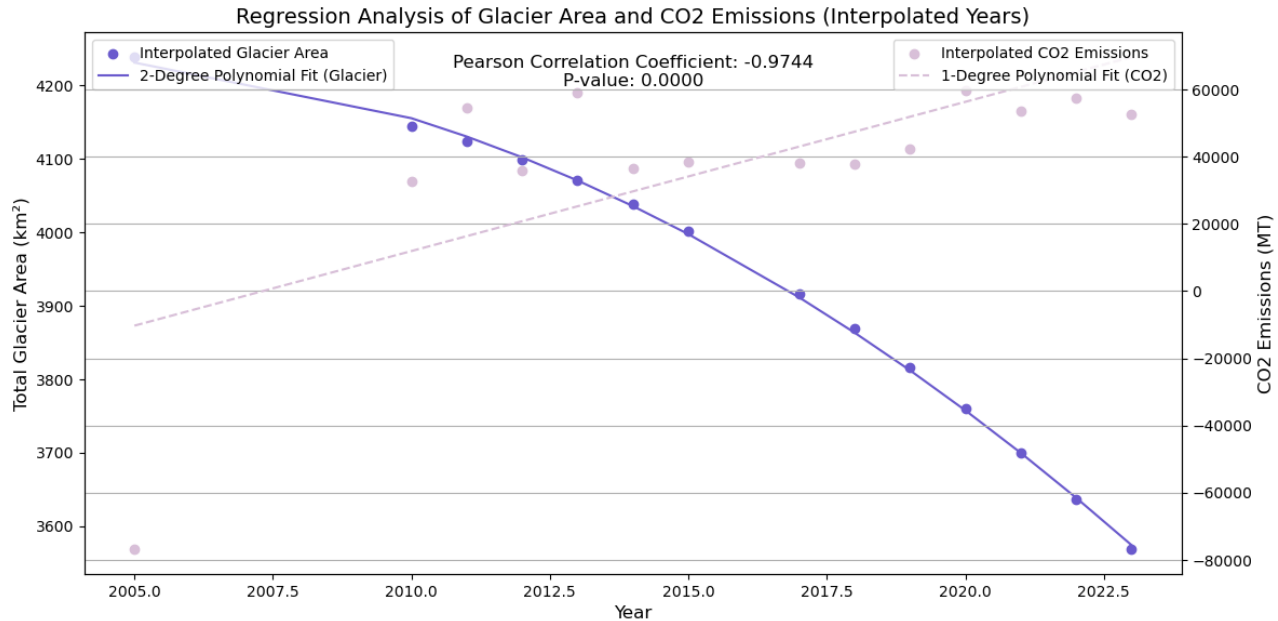


Figure 4: The negative relationship.

3 Notations

The core symbols and their definitions used in this study are summarized in Table 1, providing an overview of the key parameters and their related meanings.

Table 1: Notations used in this literature

Symbol	Description
W_j	Government revenue from tourism industry

W_l	Total government revenue of six attractions in Los Angeles
F_j	Carbon emission of Juneau
F_l	Total Carbon emission of six attractions in Los Angeles
f_h	Carbon emission of Hollywood in Los Angeles
f_d	Carbon emission of Disneyland in Los Angeles
f_g	Carbon emission of Griffith Observatory in Los Angeles
f_t	Carbon emission of America Tropical Interpretive Center in Los Angeles
f_a	Carbon emission of Avila Adobe in Los Angeles
f_c	Carbon emission of Chinese American Museum in Los Angeles
I_j	Infrastructure Maintenance Investment in Juneau
I_l	Infrastructure Maintenance Investment in Los Angeles
I_1	Infrastructure Maintenance Investment of three popular attractions in Los Angeles
I_2	Infrastructure Maintenance Investment of three unpopular attractions in Los Angeles
S_j	Resident social satisfaction of government in Juneau
S_l	Resident social satisfaction of government in Los Angeles
V_j	Yearly visitors in Juneau
V_l	Yearly visitors in Los Angeles
V_1	Yearly visitors of three popular attractions in Los Angeles
V_2	Yearly visitors of three unpopular attractions in Los Angeles
V_h	Yearly visitors in Hollywood
V_d	Yearly visitors in Disneyland
V_g	Yearly visitors in Griffith Observatory
V_t	Yearly visitors in America Tropical Interpretive Center
V_a	Yearly visitors in Avila Adobe
V_c	Yearly visitors in Chinese American Museum
E	Extra revenue in Juneau

4 Optimization & Evaluation Model of Q1

4.1 Problem Analysis

There are different dimensions and targets for us to consider in building a sustainable model. **From an economic perspective**, we need to maximize the government revenue from tourism industry, including direct revenue (Tickets, hotel accommodation fee and catering industry) and indirect revenue (potential employment opportunity). **From environic perspective**, we need to ensure the sustainable development of local environment. **From society perspective**, we need to maximize resident welfare (life quality), which is connected to their satisfaction to government. At the same time, we also need to control the increasing investment spending and prevent the tight budget among government finance. Those objectives and constraints are ebb and flow, together constitute the system operation strategy choice.

In the part of regulation effect evaluation, we need to build a **multi-level evaluation model**, comprehensively assess the the effect of adjustment measures. Concurrently, a well-designed evaluation system should also be built to adjust the optimal game strategy in time according to the evaluation results. As the case study needs to be sustainable, the optimization model and evaluation model in Juneau should be integrated organically.

4.2 Model Preparation

We use different regression formula for model preparation. Figure 5 below is the visualization of data we collect for the optimization model.

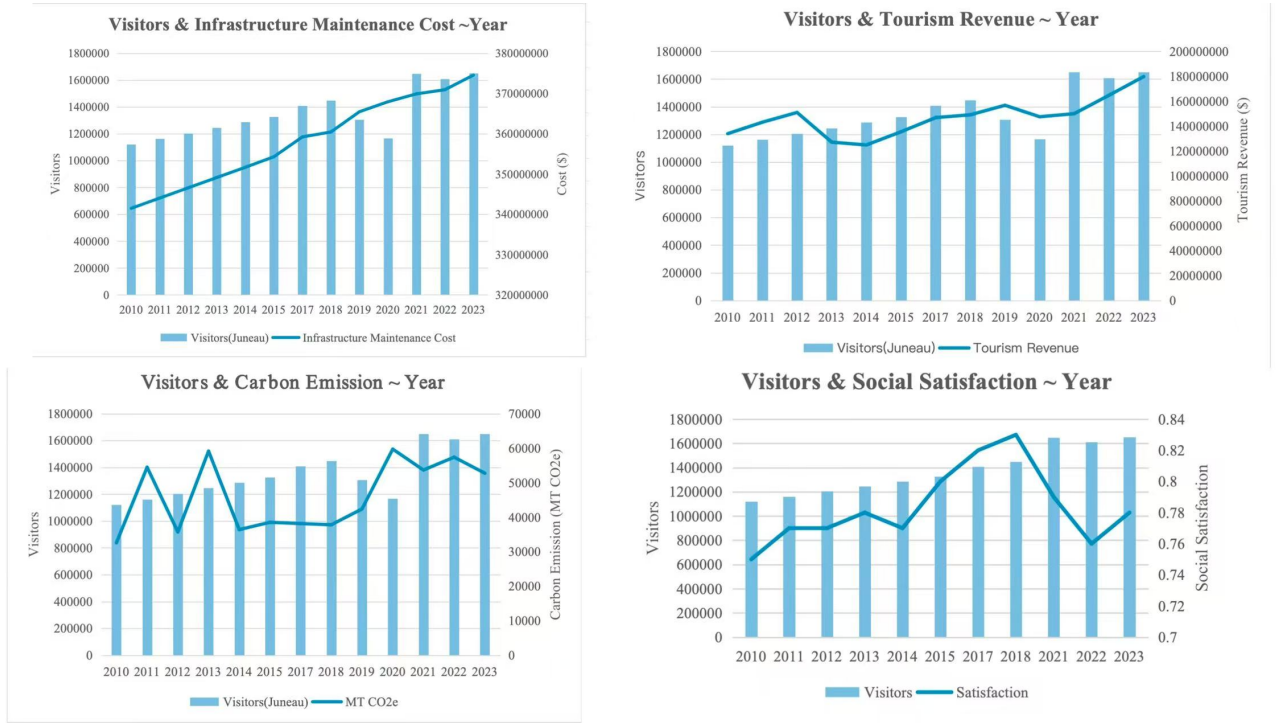


Figure 5: Data visualization [5–8]

We aim to find the best fitting curve for the Secondary axis (I_j, F_j, W_j, S_j) in each sub-figure above. We use hyperbolic function fitting, cubic function fitting, linear regression and logarithmic function regression for I_j, F_j, W_j and S_j respectively. Here are the estimated fitting functions.

$$\hat{I}_j = 53.5612v + \frac{1.33 \times 10^{13}}{v} + 2.756 \times 10^8$$

$$\hat{F}_j = 1.243 \times 10^{-13}v^3 - 3.637 \times 10^{-7}v^2 + 0.295 \times v + 6.505 \times 10^{-7}$$

$$\hat{W}_j = 52v + 76502404$$

$$\hat{S}_j = -12.337 \times \ln(1 + v) - 3.921 \times 10^{-12}v^2 + 1.992 \times 10^{-5}v + 155.198$$

In addition, by using the basic principles of macroeconomics, the **Tourism Multiplier Effect (TME)** we need the linear regression model between **Extra Revenue** and **visitors**. The estimated function is $\hat{E} = 20.1741 \times V_j + 17883749.1410$

4.3 Model Construction

4.3.1 Model Explanation

The figure below is the flowchart of our model in Problem 1. oo

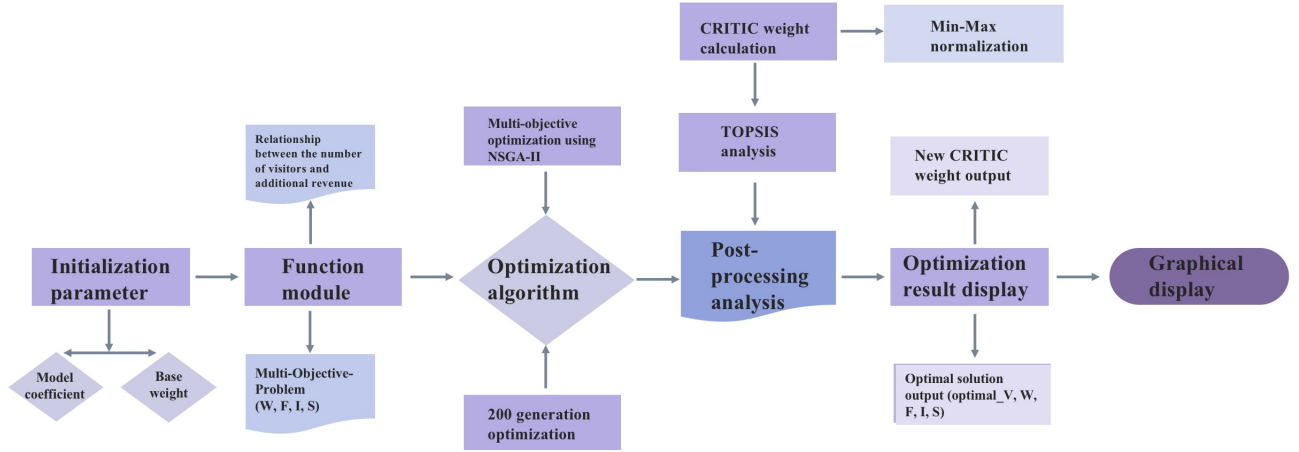


Figure 6: Flowchart of model in Q1

The model is a prior-condition multi-objective optimization with CA-TOPSIS as its evaluation model. To echo the purpose of **sustainability**, we set up a sound feedback mechanism to adjust the weight distribution of basic revenue of tourism industry and extra revenue in other aspects in a timely manner according to the TOPSIS evaluation results. The biggest highlight of the model in Problem 1 is that evaluation model and optimization model are carried out alternately.

- 1. **Prior-condition: optimization and evaluation model without EXTRA REVENUE(E):** Since all the parameters in the model is connected with V_j , we could figure out the solution of $\min I_j, \min F_j, \max W_j, \max S_j$ by using our first-stage optimization model of **NSGA-II** and **Pareto Front Model**. Besides, we use the weights we get from CRITIC in further optimization.

- **The constraint of each objective function:**

$V_j \in [1120324, 11650149]$: The lower bound and upper bound of the amount of visitors in the decade.

$F_j \leq 60000$: The highest carbon emission data is 59172, we set a bar of 60000, which means the carbon emission should be restrained, ensuring it cannot exceed 60000.

$S_j \geq 0.738$: Since we need to make sure the social satisfaction would not be too low, we set a lower bound of 0.738 which is slightly lower than the smallest score in the decade.

$300000000 \leq I_j \leq 4500000000$: There is an average ratio between I_j and W_j based on historical data, which is on average **2.5**. Meanwhile the mean of government revenue on

tourism industry is 150000000. Target on expected economic growth and market demand analysis, we set the range of I_j as $150000000 \times (2.5 \pm 0.5)$.

– **Using CRITIC to find the most realistic ratio:**

Since w_1, w_2, w_3, w_4 are four weights for W_j, F_j, I_j, S_j respectively, where $\sum_{i=1}^4 w_i = 1$. We could easily find the best ratio of weights of the four objective functions, which are **0.30918855, 0.17976395, 0.31932943, 0.19171808**.

- 2. Consider E, which is positive related with the amount of visitors, could be added as another condition for the objective functions. I_j, F_j, W_j, S_j change when we allocate E into them.
 - $F_j = 0.295v - 3.637 \times 10^{-7}v^2 + 1.243 \times 10^{-13}v^3 + 6.505 \times 10^{-7} - \min\{\frac{w_2 \times E}{42}, 5000\}$: $w_2 \times E$ means the proportion of extra revenue in reducing the carbon emission. **42** means the amount of dollars spent on reducing 1 unit of MT CO_2e in America's overall carbon market. **5000** is a threshold, revealing that the carbon emission will tend to be fixed as money is invested. (It is also an auxiliary parameter, set as 10% of total carbon footprint.
 - $I_j = 53.5612v + \frac{1.33 \times 10^{13}}{v} + 2.756 \times 10^8 - w_3 \times E$: $w_3 \times E$ means we subtract a proportion of investment, covering some of the infrastructure costs.
 - $S_j = -12.337 \times \ln(1 + v) - 3.921 \times 10^{-12}v^2 + 155.198 + 1.992 \times 10^{-5}v + \frac{w_4 \times E}{28000000}$: $\frac{w_4 \times E}{28000000}$ means the proportion of extra revenue on improving the residents' satisfaction. 28000000 is another auxiliary parameter.
- 3. After fitting four new objective functions, we can update our second-stage Pareto Front. We use the innovative algorithm, **CA-TOPSIS**, to renew w_1, w_2, w_3, w_4 continuously and find the best-score solution in TOPSIS. Here I list out the **ideal-distance matrix** (end the algorithm) .

$$A = \begin{bmatrix} 0.19 & 0.33 & 0.43 & 0.38 & 0.42 & 0.43 & 0.18 & 0.20 & 0.36 & 0.28 \\ 0.40 & 0.35 & 0.40 & 0.39 & 0.30 & 0.21 & 0.39 & 0.18 & 0.18 & 0.39 \\ 0.40 & 0.17 & 0.41 & 0.16 & 0.40 & 0.19 & 0.37 & 0.42 & 0.27 & 0.43 \\ 0.38 & 0.24 & 0.38 & 0.19 & 0.40 & 0.39 & 0.41 & 0.30 & 0.25 & 0.16 \\ 0.22 & 0.41 & 0.16 & 0.40 & 0.43 & 0.39 & 0.16 & 0.42 & 0.19 & 0.39 \\ 0.39 & 0.40 & 0.39 & 0.16 & 0.15 & 0.15 & 0.34 & 0.40 & 0.42 & 0.38 \\ 0.15 & 0.39 & 0.39 & 0.32 & 0.39 & 0.33 & 0.42 & 0.37 & 0.15 & 0.39 \\ 0.42 & 0.16 & 0.37 & 0.37 & 0.16 & 0.41 & 0.16 & 0.17 & 0.41 & 0.38 \\ 0.42 & 0.15 & 0.15 & 0.42 & 0.16 & 0.35 & 0.17 & 0.23 & 0.15 & 0.42 \\ 0.26 & 0.39 & 0.16 & 0.32 & 0.41 & 0.39 & 0.31 & 0.40 & 0.15 & 0.40 \end{bmatrix}$$

4.3.2 Pseudo Code

Algorithm 1 Multi-Objective Optimization and TOPSIS Analysis for the second stage

Require: coefficients of the fitted function and the weights

- 1: **Define function** calculate $E(v)$:
- 2: $E \leftarrow 20.1741 \cdot v + 17883749.1410$

Ensure: Best solution based on TOPSIS analysis

- 3: **Initialize Multi-Objective Problem:**

```

4:   $n\_var \leftarrow 1, n\_obj \leftarrow 4, ieq\_constr \leftarrow 4, xl \leftarrow [1120324], xu \leftarrow [1650149]$ 
5:  Define evaluate function:
6:   $E \leftarrow [\text{calculate } E(v_i) \text{ for each } v_i \text{ in } v]$ 
7:   $W_j \leftarrow 52 \cdot v + 76502404$ 
8:   $F_j \leftarrow 0.295 \cdot v - 3.637 \cdot 10^{-7} \cdot v^2 + 1.243 \cdot 10^{-13} \cdot v^3 + 6.505 \cdot 10^{-7} - \min(w_2 \cdot (E/42), 5000)$ 
9:   $I_j \leftarrow 53.5612 \cdot v + 1.33 \cdot 10^{13} \cdot (1/v) + 2.756 \cdot 10^8 - w_3 \cdot E$ 
10:  $S_j \leftarrow -12.337 \cdot \ln(1 + v) - 3.921 \cdot 10^{-12} \cdot v^2 + 155.198 + 1.992 \cdot 10^{-5} \cdot v + w_4 \cdot E / 2.8 \cdot 10^7$ 
11:  $out['F'] \leftarrow [-W, F, I, -S]$ 
12:  $g \leftarrow [F - 60000, I - 450000000, 0.738 - S, 300000000 - I]$ 
13:  $out['G'] \leftarrow g$ 
14: Run Optimization:
15:  $algorithm \leftarrow \text{NSGA2}(pop\_size = 100)$ 
16:  $res \leftarrow \text{minimize}(problem, algorithm, ("n\_gen", 200), seed = 42, verbose = True)$ 
17: if  $res.F \neq \text{None}$  and  $\text{len}(res.F) > 0$  then
18:    $new\_weights \leftarrow \text{critic\_weight\_calculation}(objectives)$ 
19:    $w_1, w_2, w_3, w_4 \leftarrow new\_weights$ 
20:    $normalized\_objectives \leftarrow \text{MinMaxScaler}().fit\_transform(objectives)$ 
21:   TOPSIS analysis to find best index with highest TOPSIS scores
22:   Output:
23:   Print optimal values:  $optimal\_V, W, F, I, S$ 
24: end if=0

```

4.4 Model Solution

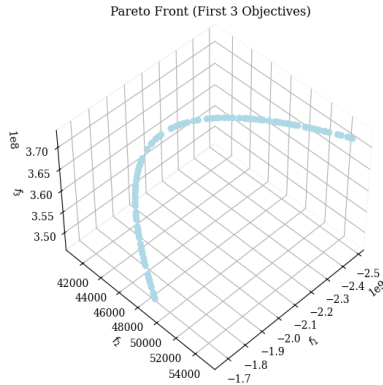


Figure 7: First-stage Pareto Front

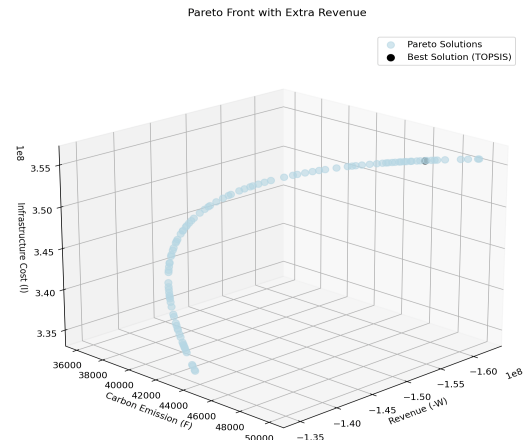


Figure 8: Second-stage Pareto Front

There are two main optimizing part in Problem 1 (basic revenue in tourism & extra revenue). The optimal solution in first-stage Pareto Front without extra revenue (with only basic revenue) is:

$$V_j = 1175225, W_j = 137614091.32, F_j = 46125.34, I_j = 349863432.41, S_j = 0.76$$

Using CRITIC we could determine w_1, w_2, w_3, w_4 as **weights** for the second-stage Pareto Front.

Since governments usually use extra revenue in indicators above to improve people's livelihood. We need to focus more on the optimal solution of the second-stage Pareto Front (with extra revenue),

which is:

$$V_j = 1621472, W_j = 176462472.32, F_j = 47011.35, I_j = 354493766.45, S_j = 0.8185$$

with the weight ratio of extra revenue as

$$w_1 = 0.1503, w_2 = 0.3635, w_3 = 0.1507, w_4 = 0.3356$$

The data above is optimized in a certain scope: On the premise that the number of tourists has remained almost unchanged, carbon emissions and infrastructure investment have been significantly reduced compared with the data of the past three years, while ensuring the steady growth of government tourism revenue. The figures above visualize the best solution in **two stages 3D-Pareto Front**.

4.5 Sensitivity Analysis

Since the sensitivity analysis needs to analyse different fitting coefficients, we recall the **initial fitting function**: hyperbolic function fitting ($\hat{I}_j = a_i \cdot v + b_i \cdot \frac{1}{v} + c_i$), cubic function fitting ($\hat{F}_j = a_f \cdot v + b_f \cdot v^2 + c_f \cdot v^3 + d_f$), linear regression ($\hat{W}_j = a_w \cdot v + b_w$) and logarithmic function regression ($\hat{S}_j = a_s \cdot \ln(1 + v) + b_s \cdot v^2 + c_s + d_s \cdot v$). For each coefficient, we choose a **fluctuation** of $\pm 10\%$ to observe its sensitivity.

Overall, it is clear to see that the objective function of tourism revenue and infrastructure maintenance investment are less sensitive to changes of their model coefficient, while carbon emission is more sensitive, and residence satisfaction is extremely sensitive to coefficient variations and small deviations. This indicates that minor changes in the parameters affecting residence satisfaction can lead to significant impacts on the overall model outcomes, highlighting the need for careful consideration of these factors. Figures below visualize the general information of sensitivity test on each coefficient and the changing Pareto Front.

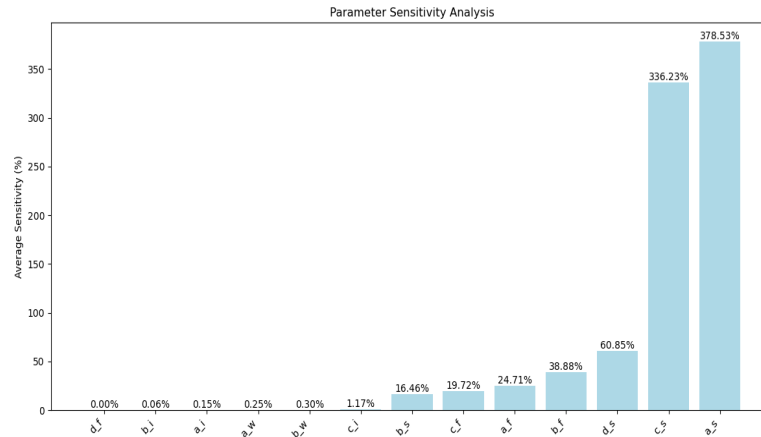


Figure 9: Coefficients Sensitivity in Problem 1

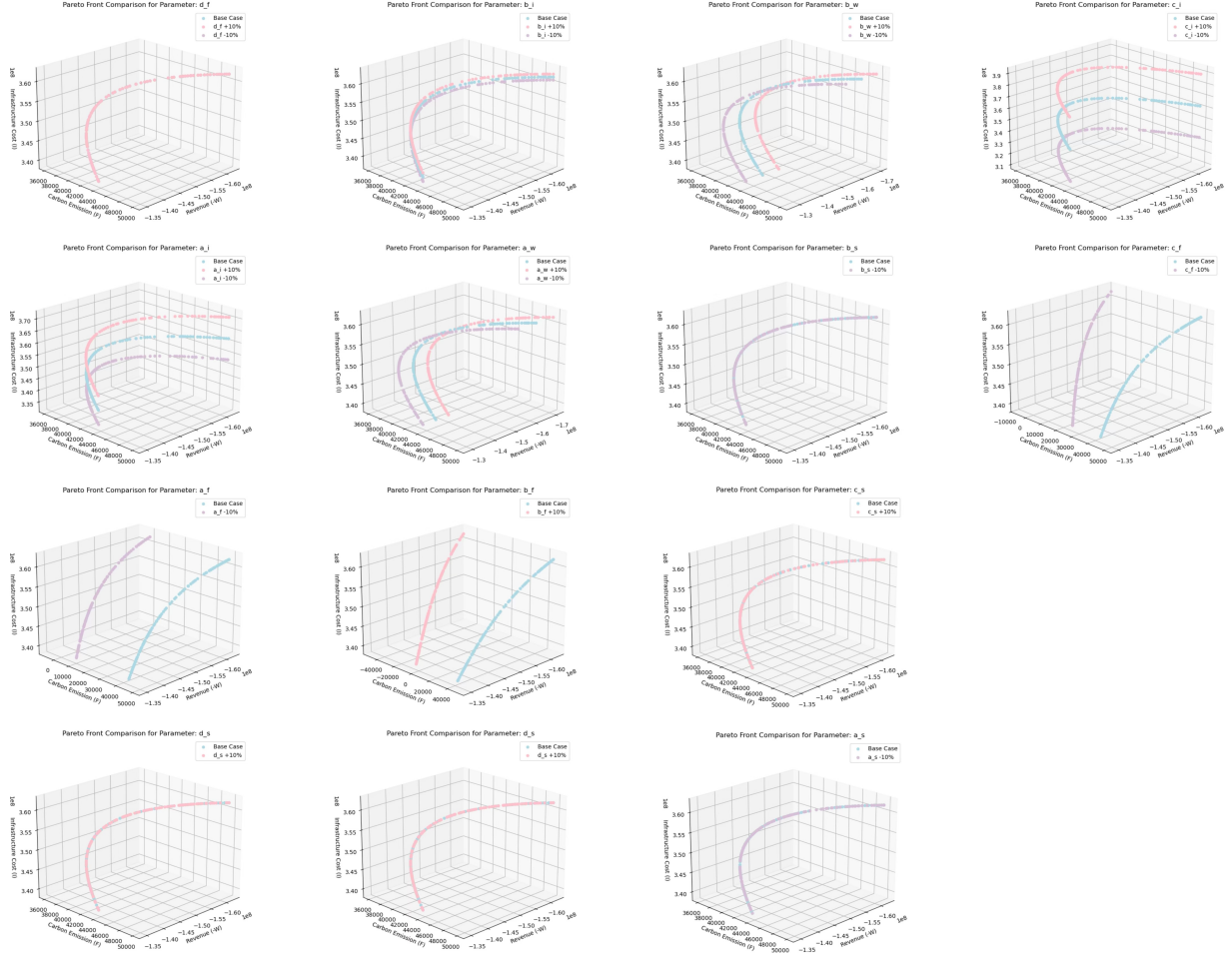


Figure 10: Change of Pareto Fronts among coefficients in Problem 1

5 Optimization & Evaluation Model of Q2

5.1 Problem Analysis

In order to solve the problem of how to **balance** the destinations affected by overtourism and promote the less visited attractions, we **selected six tourist attractions**: Hollywood Universal Studio, Disneyland, Griffith Observatory, America Tropical Interpretive Center, Avila Adobe Museum and Chinese American Museum. Then we **classified** them into two sections, the first three are popular attractions, the last three are unpopular attractions. Our goal is to realize the **diversion** of popular scenic spots to unpopular scenic spots on the premise of ensuring the optimization of the objective function.

There are different dimensions and targets for us to consider in building a sustainable model. Similar with Q1, we need to maximize the government revenue from tourism industry (W_I); ensure the sustainable development of local environment (F_I); maximize resident welfare (S_I) and control the increasing investment on infrastructure maintenance (I_I). In each objective function, we refine it into

the corresponding summation of six attractions (e.g. $F_l = f_h + f_d + f_g + f_t + f_a + f_c$, etc.). These objectives and constraints are ebb and flow, together constitute the system operation strategy choice.

In the part of regulation effect evaluation, we need to build a **multi-level evaluation model**, comprehensively assess the the effect of adjustment measures. Also, in order to achieve diversion, the government can reduce the **consumption tax** on unpopular scenic spots (to reduce the cost of tourists to travel to unpopular scenic spots) and increase **infrastructure investment**. To increase the guiding significance of the model, the optimization model and evaluation model in Juneau should be integrated organically

5.2 Model Preparation

Since historical data [9] on the satisfaction of residents near each attraction and infrastructure maintenance investment of each attraction are missing, we need to build up data-simulation models. According to the first type of data, we develop an **ordinary differential equation** model (A stable model comparing to ordinary curves) related to tourist carrying capacity, ensuring that the residents will not be satisfied with too many or too few visitors. Here, we sum up the visitor capacity data of the 6 destinations we choose in Los Angeles, annual visitor volume for different attractions and annual resident satisfaction in LA to fit the determination coefficient. [10–14]

$$\left\{ \begin{array}{l} \left\{ \begin{array}{l} \frac{dS}{dV} = a(V - V_{opt}) + b \\ S(V_{opt}) = c \\ V = \sum_{i=h,d,f,t,a,c} V_i \\ V_{opt} = \sum_{i=h,d,f,t,a,c} V_{opti} \end{array} \right. \\ \text{where } \left\{ \begin{array}{l} a = -0.643 \cdot 10^{-15} \\ b = -1.59 \cdot 10^{-8} \\ c = 0.54329 \end{array} \right. \end{array} \right.$$

According to the second type of data, we develop a direct scale model to find the relationship between the yearly infrastructure investment on three specific popular scenic spots (Disneyland, Hollywood and Griffith Observatory), three specific less-visited scenic spots and the total infrastructure investment of LA.

$$I_1 = I_l \cdot \frac{V_1}{V_l}$$

$$I_2 = I_l \cdot \frac{V_2}{V_l}$$

We use different regression formula for model preparation. We need to identify the best-fitting curve for the carbon emissions associated with six destinations based on their yearly visitor numbers. [6] They are as follows:

$$\hat{f}_h = 1.932 \cdot 10^5 + 0.0033 \cdot V_h$$

$$\begin{aligned}\hat{f}_d &= -5381 + 0.0268 \cdot V_d \\ \hat{f}_g &= 1.444 \cdot 10^{-7} + 0.0978 \cdot V_f - 3.064 \cdot 10^{-8} \cdot V_f^2 - 1.64 \cdot 10^{-15} \cdot V_f^3 \\ \hat{f}_t &= 7338.16 \cdot \ln(V_t) - 60581.25 \\ \hat{f}_a &= 11989.43 + 0.166 \cdot V_a \\ \hat{f}_c &= 159.314 + 0.1693 \cdot V_c\end{aligned}$$

We could clearly see the trend of visitors amount and carbon emissions over the years in the figure below, which indicate the six fitting formulas above.

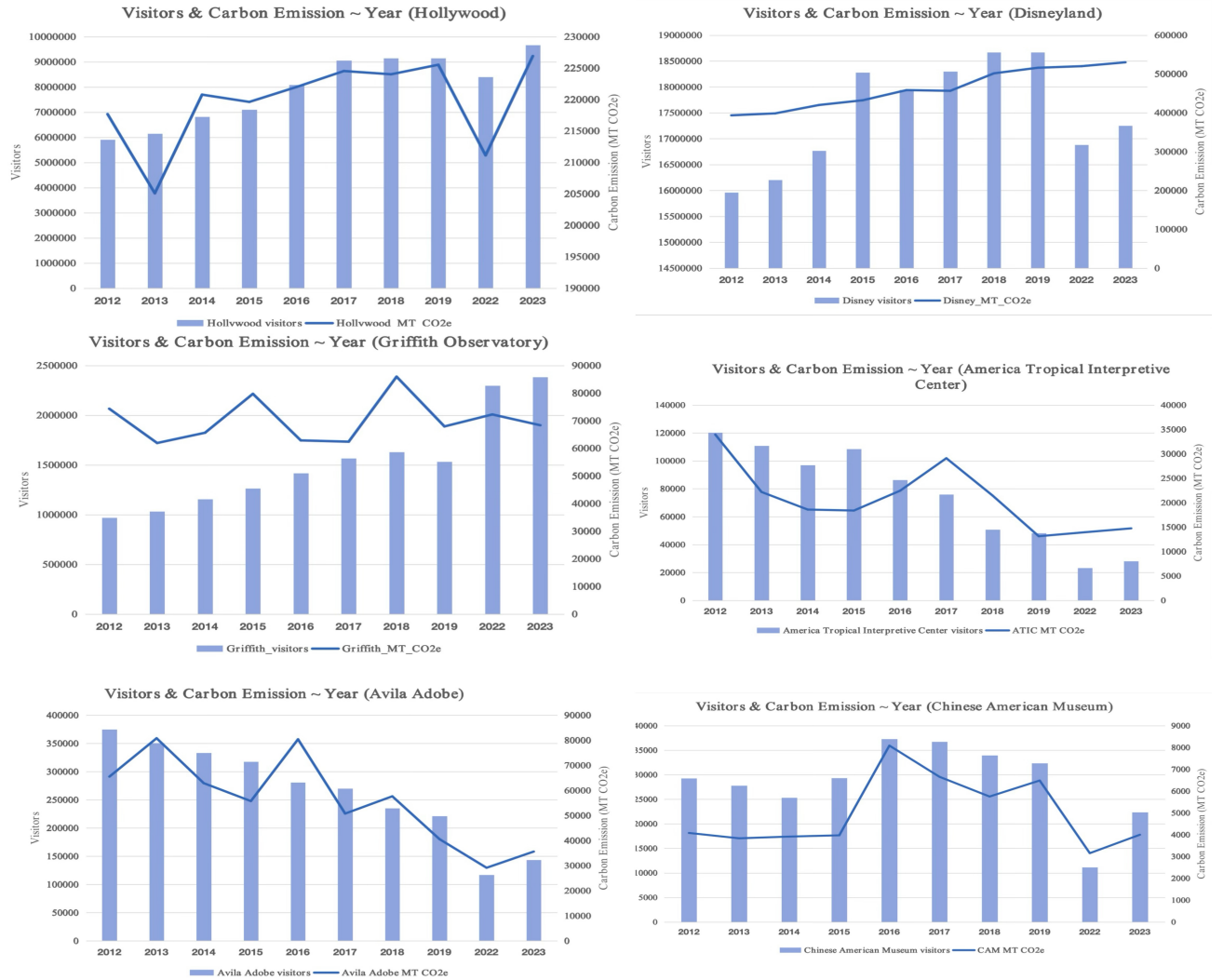


Figure 11: Data visualization [6]

5.3 Model Construction

The figure below is the flowchart of our model in Problem 2.

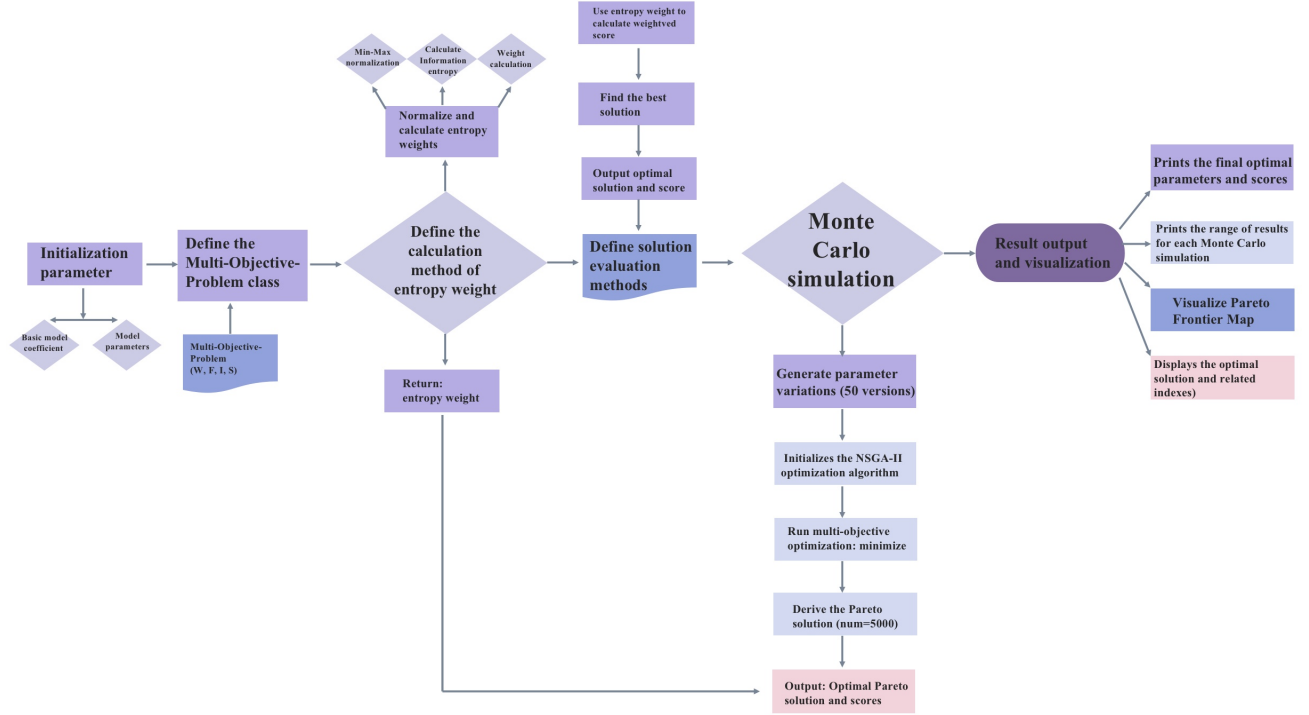


Figure 12: Flowchart of model in Q2

5.3.1 Model Explanation

We designed a **EMCM** model (Monte Carlo simulated multi-objective optimization model evaluated by Entropy Weight Method). The model is a **multi-objective** optimization framework incorporating **entropy weight** analysis to **dynamically evaluate** objective importance. Using NSGA-II, it identifies the **Pareto optimal solutions** and integrates entropy-based reweighting for comprehensive decision-making. The highlight of the model lies in its **adaptive weighting mechanism**, which ensures balanced optimization across to-determined policies.

- **1. Evaluation weight derivation:** The NASG-II algorithm is run on the basic model once, and the points on the Pareto front are analyzed by Entropy Weight Method to find out the weights w_1, w_2, w_3, w_4 . Then define the weighted-sum-evaluate function as $Z = w_1 \cdot (-\frac{W_I}{W_{\max}}) + w_2 \cdot (\frac{F_I}{F_{\max}}) + w_3 \cdot (\frac{I_I}{I_{\max}}) + w_4 \cdot (\frac{S_I}{S_{\max}})$, the function will be used in the Monte Carlo simulated multi-objective optimization model to investigate the reference optimal solution on the Pareto front.
 - **Using Entropy Weight Method:** The **robustness** is stronger than the CRITIC method and will perform better in Monte Carlo stochastic simulations
- **2. Monte Carlo simulation:**
 - Number of iterations = 50, **floating scale** of the parameters to be varied = 0.2
Floating indicators should be defined by the government, increasing tourism tax for popular scenic spots to limit the flow of tourists, and appropriately reducing tourism tax for unpopular scenic spots to promote the growth of tourist numbers, so as to achieve the purpose of

diversion. Increase infrastructure investment for less popular attractions, thereby increasing the attraction of less popular attractions to tourists. We choose 0.2 here to be the floating scale of the parameter.

- In each Monte Carlo cycle, NASG-II algorithm is run to obtain a set of Pareto frontier solutions, all solutions in the solution set are **scored** by weighted sum method, the optimal solution (with the smallest value) is found, and the value of the objective function and decision variable, as well as the value of each floating index in the optimal case are obtained.

5.3.2 Pseudo Code

Algorithm 2 Multi-Objective Optimization and Monte Carlo Simulation Analysis

Require: base model coefficients and parameters to be varied

```

1: Define class MultiObjectiveProblem(params):
2: Initialize: basic information of the optimization problem
3: Define evaluate function:
4:   Extract decision variables v1, v2, v3, v4, v5, v6 and representations of W, F, I, S
5:   out['F'] ← [-W, F, I, -S]
6:   g ← [constraints]
7:   out['G'] ← g
8: Define function run_optimization(params):
9:   Return minimize(problem, algorithm, ('n_gen', 200), seed=42, verbose=False)
10: monte_carlo_simulation(base_params, n_samples=50, variation=0.2):
11:   Initialize best_score, best_params, objective tracking list
12:   Define function aggregate_obj(values):
13:     coefficients ← [0.3372, 0.2091, 0.2882, 0.1655] # derived by Entropy Weight Method
14:     Return sum(values·coefficients)
15:   i = 1 to 50 do:
16:     Generate test_params with variations
17:     res ← run_optimization(test_params)
18:     If res.F is not None and len(res.F) > 0:
19:       Store objective values
20:       aggregated ← [aggregated_obj(ind) for ind in res.F]
21:       best_score ← max(aggregated[argmin(aggregated)], best_score)
22:       test_params.copy
23:   Return best_params, best_score
24:   base_results ← run_optimization(base_params)
25: Output:
26:   optimal parameters and solution values from Monte Carlo simulation =0

```

5.4 Model Solution

5.4.1 The Solution of the Objective Function

i.) According to the weighted sum method, the optimal evaluation is decomposed as follows:
Optimal solution values:

- **Objectives:**

- W_l (Revenue): 12,436,306,530.37
- F_l (Carbon Emission): 900,894.82
- I_l (Infrastructure Cost): 929,292,490.84
- S_l (Satisfaction): 0.5465

- **Decision Variables:**

- V_h : 9,659,979.36
- V_d : 18,665,767.29
- V_g : 2,382,589.79
- V_t : 362,732.74
- V_a : 498,907.35
- V_c : 92,910.34

iii.) According to the Monte Carlo simulation results, the ranges of the four objective functions are as follows:

- Revenue (W_l): Min: 9,085,507,347.46 ; Max: 12,482,201,239.61
- Carbon Emission (F_l): Min: 722,673.72 ; Max: 936,307.45
- Infrastructure Cost (I_l): Min: 900,000,101.18 ; Max: 1,154,214,966.40
- Satisfaction (S_l): Min: 0.5300 ; Max: 0.5536

5.4.2 The Optimized Floating Scale

On the basis of obtaining the optimal solution, the floating coefficient of each variable is obtained as follows: **Optimal parameters from Monte Carlo simulation:**

- c_h (Average consumption of tourists in Hollywood): 354.4419 (change from base: +18.15%)
- c_d (Average consumption of tourists in Disneyland): 458.0708 (change from base: +14.52%)
- c_g (Average consumption of tourists in Griffith Observatory): 158.7238 (change from base: +5.82%)
- c_t (Average consumption of tourists in Tropical Interpretive Center): 90.1589 (change from base: -9.84%)

- c_a (Average consumption of tourists in Avila Adobe): 84.3290 (change from base: -15.67%)
- c_c (Average consumption of tourists in Chinese American Museum): 99.1649 (change from base: -0.84%)
- I_1 (Government infrastructure investment in the first three attractions): 29.2697 (change from base: -4.62%)
- I_2 (Government infrastructure investment in the latter three attractions): 31.9204 (change from base: +4.02%)

5.4.3 Interpretation of Data

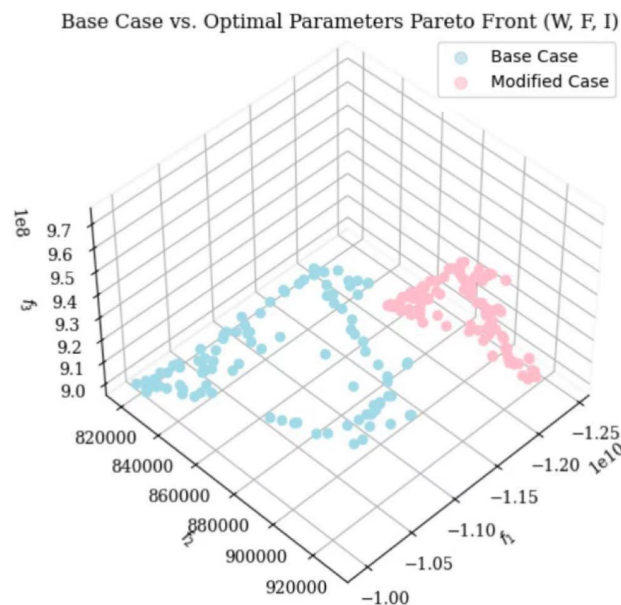


Figure 13: First & Second Stage Pareto Front

i.) Comparative analysis of optimal Pareto frontier solutions and historical data Compared to historical data, tourism revenue increased by approximately 11.9%, carbon emissions increased by 16%, and changes in infrastructure investment and satisfaction were within normal fluctuations. From this perspective, the development direction of the optimization model **aligns** with the coefficients obtained through the entropy weight method, indicating that our optimization model is **theoretically self-consistent** and possesses scientific and practical significance.

ii.) The floating ratio is in line with expectations According to the optimization results, the floating ratios of parameters show that c_h , c_d , and c_g are in positive growth while c_t , c_a , and c_c exhibit negative growth. This **aligns** with the government's strategy to adjust the tourism taxes for both popular and less-visited scenic spots within the optimization model. Additionally, the negative growth of I_1 and positive growth of I_2 further reflects the government's projected infrastructure expenditure in the optimization framework. The results demonstrate that diversion and optimization can occur **simultaneously**, achieving the expected outcomes of our optimization model.

iii.) **Comparative analysis of optimal Pareto frontier solutions and base data** Relative to the base data before adjustments, income increased by 15.0%, carbon emissions decreased by 0.1%, infrastructure investment fell by 4.4%, and satisfaction decreased by approximately 0.06%. The reduced dispersion of the optimized Pareto front indicates a superior effect of multi-objective optimization. This suggests that the parameter adjustments have positively impacted the optimization model, successfully contributing to the real-world objectives of shunting.

5.5 Sensitivity Analysis

We processed each group of parameters with an offset degree of plus or minus 10%, comparing the Pareto front before and after the offset. The differences were ranked, with the approximate ranking **aligning closely** with the weight ranking derived from information entropy ($W > I > F > S$).

Furthermore, we discovered that parameter changes affecting Los Angeles resulted in a much smaller shift compared to Juneau. This can be attributed to the larger resident and tourist population in Los Angeles, which leads to slower regulatory effects from the government. Nevertheless, parameters related to revenue (W) and satisfaction (S) continue to be significant factors influencing the optimization results.

In conclusion, the government can facilitate the diversion between popular and less-visited scenic spots through the regulation of tourism taxes and infrastructure investment.

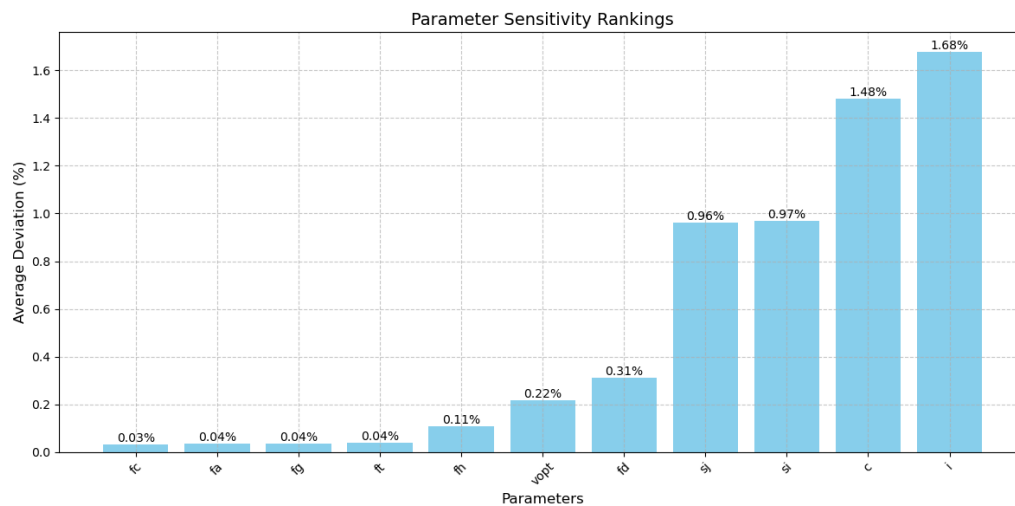


Figure 14: Coefficients Sensitivity in Problem 2

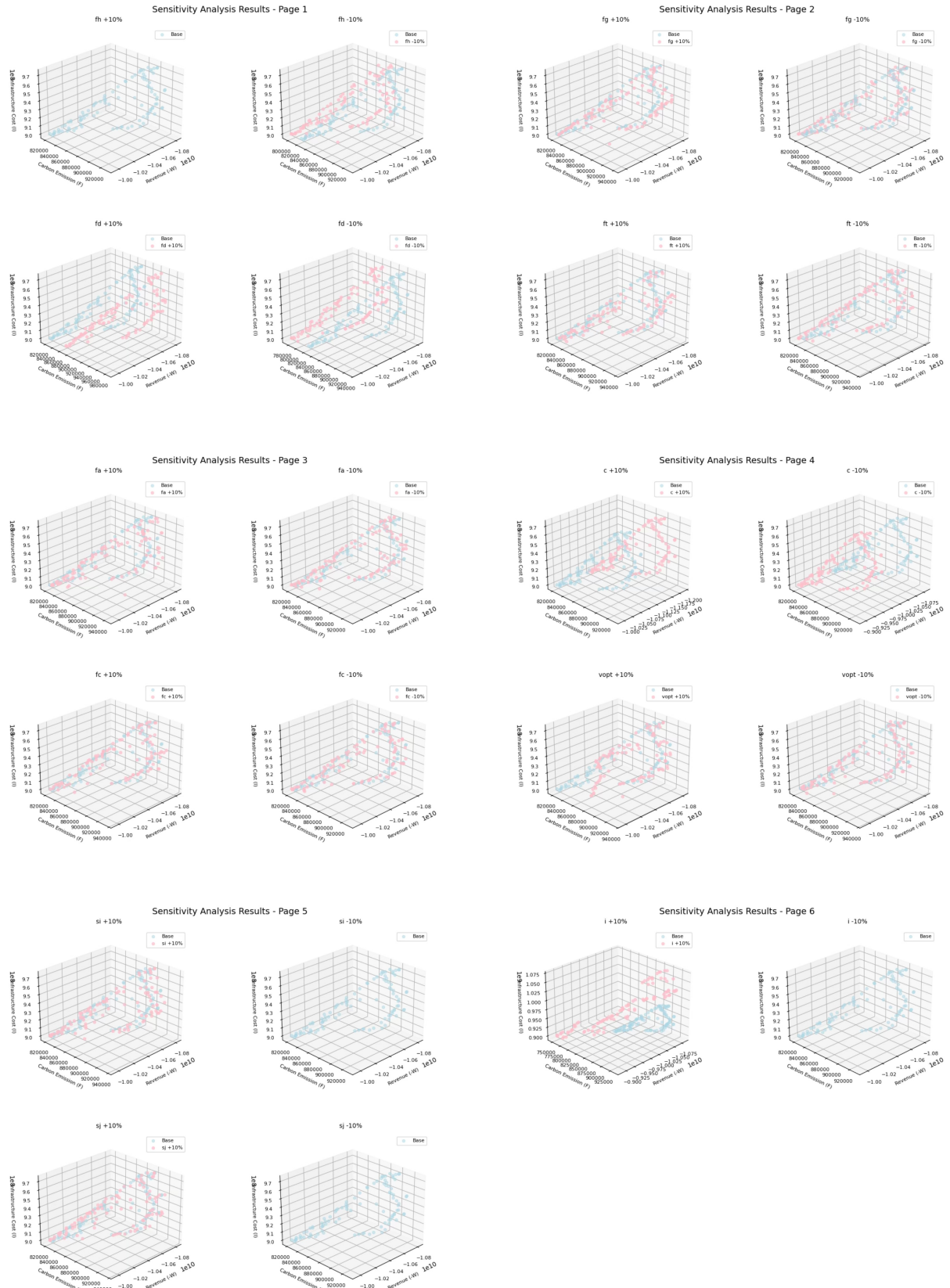


Figure 15: Change of Pareto Fronts among coefficients in Problem 2

6 Model Assessment and Promotion

6.1 Model Assessment

The core idea of our code is using the **hybrid model** with alternative evaluation and optimization (two stages). We select four **representative indicators**, ensuring the objective functions are optimized based on our modeling idea.

Advantages

The main concept of **sustainable development** is well demonstrated by our model. Here are the comprehension from different perspectives.

- By optimizing the distribution of tourists across both popular and less-visited attractions, our model offer suggestions of passenger traffic change directions to local governments. Such an operation management enlightens citizens and decision makers to determine more appropriate choices on dispatch.
- Our model formulate intelligent decisions about the allocation of additional government revenue. We optimize the **ratio rate** of different objective functions, ensuring the maximization of benefit among government, residents and visitors and a virtuous circle of government finance.

Since our model is implemented both in popular tourist-oriented micropolis and metropolitans with developed entertainment industry, the parameters and range of restrictions are robust to be applied in distinct types of cities.

Limitations

Meanwhile, our model has some limitations due to the lack of data and technology. Since our model needs some target data which are hard to quantified, we may suffer higher error and biases.

- Most of the parameters in our model is fitted based on 10-year period. The regression model of some objective functions may be overfitting or underfitting. Such troubles might cause low generalization ability and model performance.
- Some unavoidable data in our model might be hard for government to capture (the carbon emission produced by visitors in some attractions etc.). We need to bring data into the existing model and simulate an approximate value, which might cause potential error.

6.2 Future Work

Model extension & Application

There are several ways to extend our optimization-evaluation model. One way is to use machine learning model instead of regression by stretching the timeline and training labeled data. Since there are plenty of monthly statistical data, another way is to use time-series model to deal with periodic data. Both of the advanced method will receive high precision predicted value.

Our model could be widely used in business development planning and government macro forecasting model, especially in the areas of **policy evaluation and determination**. In summary, the versatility of our model allows it to be used in various situation for smart city, ultimately contributing to more sustainable tourism practices and improve the public satisfaction and government credibility.

7 Memo to the tourism council

Analysis and Suggestions on Sustainable Tourism Development in Juneau

Dear Members of the Tourism Committee:

Based on in-depth analysis of Juneau's tourism industry in previous years, our team completed a series of modeling and optimization efforts to support the sustainable development of Juneau's tourism industry. Here are our main conclusions and recommendations.

Juneau received more than 1.6 million visitors in 2023, and simulations predict that the number of tourists will continue to increase, resulting in: infrastructure overload, increased carbon emissions (strongly correlated with glacier degradation, which could make Juneau less attractive to tourists), increased environmental degradation, and decreased life satisfaction among residents. According to the data of previous years, taking the number of tourists as a variable, four objective functions are fitted out, which are: **maximize total visitor consumption (W), minimize tourist carbon footprint (F), minimize investment in infrastructure maintenance and construction (I), and maximize satisfaction of local residents (S).**

We extend the model to other overtourism areas to verify the stability of the model. Based on analysis, we give the following suggestions:

Manage the Flow of Tourists: Our model suggests a theoretical optimal number of tourists. The government can limit daily visitors through a reservation system and implement a floating tax rate to control tourist numbers when they exceed the optimal value.

Optimize Visitor Distribution: Juneau scenic spots can be divided into popular scenic spots and unpopular scenic spots. The government can increase promotion of unpopular scenic spots, using our model to guide tourists from popular attractions to unpopular ones.

Environmental Protection and Testing: The government can establish an environmental monitoring system to monitor the changes of carbon emissions and the carrying capacity of the natural environment in real time, so as to dynamically regulate the number of tourists and realize the sustainable development of tourism. In addition, each visitor is charged a certain amount of environmental protection fees to reduce carbon emissions and maintain infrastructure.

There are several ways to optimize results.

Data Quantity and Source Problem: First of all, the data sources we collected were limited, and only 2010-2023 data were considered in the optimization model, and some data were analyzed according to the fitting model. In the further improvement of the model, if there are more quantities (such as data from 1980-2023) and more details (such as data from every month of every year), the ARIMA method can be further adopted to deal with seasonal or periodic processing of tourist data, so as to improve the practical significance of the optimized model.

Optimization of Fitted Models: Secondly, our current fitting model data are all from 2010 to 2023, and too little data leads to insufficient accuracy of the fitting model. If there is data for each year and each month, the fitted model can be further optimized to improve results.

Through scientific management and planning, we believe Juneau can achieve sustainable tourism development. Under the condition that the environmental load is not exceeded, the tourism income and residents' satisfaction are maximized, so as to achieve a win-win situation. I hope this memo can provide some reference for your committee's decision-making. If you have any further questions, we will be happy to answer them.

Team#2510156

2025.1.27

8 Conclusions

In order to realize the sustainable development of popular tourist cities, we construct a highly compatible optimization model based on NASG-II algorithm. When considering the sustainable tourism problem in Juneau, we combined the base model with a CA-TOPSIS evaluation model. Our goal is to balance and optimize the economic income, environmental sustainability, and social well-being. To be more specific, the government is suggested to consciously reduce the number of visitors per year by around 5%, thereby striking a balance between the 4 concerned objectives. We also provide a preferred weight for the government to better manage the extra revenues. This demonstrates how balanced resource allocation and feedback adjustments can achieve sustainable tourism.

When discussing the adaptability and practicality of our model, we choose one the most popular destinations in the world: Los Angeles. On the basis of NASG-II model, in order to solve the problem of tourist flow between popular and unpopular scenic spots in the city, we adjusted the parameters of the two functions with the strongest correlation with the number of tourists, and combined with Monte Carlo algorithm, obtained the optimal floating parameters and the corresponding optimal solutions within a certain floating interval. The optimal solution is determined by Entropy-Weighted-Sum-Method. It suggests that the government can slightly reduce the tourism taxes and investigate more on the infrastructure construction of the unpopular scenic spots, which can achieve the goal of promoting these attractions. However, it is noticeable that in our model, the economic benefits may be overemphasized. In actual practices, AHP method can also be taken into consideration, whose weight can be determined by the actual need of the government.

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Report on Use of AI

1. Sider

Query1: < write a sentence to express the beautiful view in Juneau, including the Mendenhall Glacier, whale watching tool and sth else.>

Output: <In Juneau, visitors are captivated by the stunning views of the Mendenhall Glacier, where the breathtaking blue ice meets the lush green forests, while the excitement of whale watching adds an unforgettable touch as majestic humpbacks breach the surface, creating a perfect harmony of nature's splendor.>

2. Sider

Query1:<Give a brief introduction of LA, 50 words>

Output: <Los Angeles, often referred to as LA, is a vibrant city in Southern California known for its entertainment industry, particularly Hollywood. It boasts diverse neighborhoods, stunning beaches, and cultural landmarks. With a Mediterranean climate, LA attracts millions of visitors annually, offering a unique blend of culture, arts, and outdoor activities.>

3. OpenAI *ChatGPT* (Nov 5, 2023 version, ChatGPT-4)

Query1:<basically tell me what it is TOPSIS>

Output: <TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution) is a method used to help make decisions when there are multiple conflicting criteria to consider. It ranks alternatives based on how close they are to an ideal solution.

Here's a simple breakdown of what it does:

- ****Ideal solution****: The best possible outcome for each criterion.
- ****Negative ideal solution****: The worst possible outcome for each criterion.
- It compares each alternative to these two solutions and calculates how "close" they are to the ideal solution and how "far" they are from the negative ideal solution.
- The alternative that is closest to the ideal and farthest from the negative ideal gets the highest rank.

In short, TOPSIS helps pick the option that is the best overall by measuring its distance from an ideal best-case scenario, while also considering the worst-case scenario.>

4. OpenAI *ChatGPT* (Nov 5, 2023 version, ChatGPT-4)

Query1:<The connection between NSGA-2 and Pareto Front>

Output: <NSGA-II (Non-dominated Sorting Genetic Algorithm II) is a popular evolutionary algorithm used for solving multi-objective optimization problems. One of the key concepts it works with is the **Pareto Front**, and the relationship between NSGA-II and the Pareto Front is central to how the algorithm performs and finds solutions.

What is the Pareto Front?

In multi-objective optimization, there are multiple conflicting objectives. The **Pareto Front** (or Pareto-optimal set) represents the set of solutions where no other solution is

strictly better across all objectives. A solution is said to be **Pareto optimal** if you cannot improve one objective without sacrificing at least one other objective.

For example, in a two-objective problem where you're trying to maximize both $f_1(x)$ and $f_2(x)$, a Pareto optimal solution is one where you cannot increase $f_1(x)$ without decreasing $f_2(x)$, or vice versa.

How Does NSGA-II Relate to the Pareto Front?

NSGA-II uses the concept of **Pareto dominance** and **non-dominated sorting** to drive its search for optimal solutions in a multi-objective space. Here's how it connects with the Pareto Front:

Non-dominated Sorting:

1. NSGA-II sorts the population of solutions into different "fronts" based on Pareto dominance.
2. A solution is said to **dominate** another solution if it is better in at least one objective and no worse in all others.
3. The first **Pareto front** (or **front 1**) consists of solutions that are not dominated by any other solutions in the population.
4. The second **Pareto front** (or **front 2**) consists of solutions that are dominated by the solutions in front 1, but not dominated by any other solutions outside of it, and so on.

Crowding Distance:

1. After sorting the population into different Pareto fronts, NSGA-II further refines the selection of solutions using **crowding distance**.
2. Crowding distance is a measure of how close a solution is to others in the objective space. The goal is to maintain diversity in the population and avoid clustering solutions too closely together, especially on the Pareto Front.

Selection for Crossover:

1. Once the population has been sorted into Pareto fronts, NSGA-II uses these fronts to select solutions for crossover and mutation operations. It prioritizes solutions from the first Pareto front (the most optimal) and then uses the crowding distance to maintain diversity.
2. This helps the algorithm explore the Pareto Front efficiently, ensuring that solutions are spread out across the front rather than converging too quickly on a single region.

The Role of NSGA-II in Approaching the Pareto Front

- NSGA-II helps generate and maintain a **diverse set of solutions** that approximate the entire Pareto Front.
- Over generations, NSGA-II progressively evolves the population toward the **true Pareto Front**, using non-dominated sorting to prioritize better solutions and crowding distance to ensure diversity.
- The **final population** ideally contains a set of solutions that form a good approximation of the Pareto Front, covering different trade-offs between the objectives.

Summary of the Connection:

- **NSGA-II** uses **Pareto dominance** to sort solutions into different **Pareto fronts**, with the first front being the most optimal.
- The algorithm then aims to maintain diversity and generate a wide spread of solutions across the **Pareto Front**.
- The end result of an NSGA-II run is a **set of solutions** that approximates the **Pareto Front** for a given multi-objective optimization problem.>

5. OpenAI *ChatGPT* (Nov 5, 2023 version, ChatGPT-4)
Query1:<Briefly tell me how to find the keywords>

Output: <To find keywords, particularly for research or content analysis, you can follow these basic steps:

Identify the Main Topic: Understand the central theme or subject matter of your text, research, or article.

Highlight Important Terms: Look for repeated words, terms, or phrases related to the topic, such as specific concepts, methods, or categories.

Use Frequency Analysis: Tools like word clouds, text analyzers, or manual frequency counts can help identify words that appear most often.

Look for Technical Terms: Focus on specific jargon, terminology, or names unique to the field.

Consider Context: Keywords should reflect the purpose of the content and resonate with the target audience or search intent.

Use SEO Tools (if applicable): Tools like Google Keyword Planner, Ahrefs, or SEMrush can help identify keywords relevant to your audience, especially for online content.

These steps will help you isolate the most relevant and significant keywords for your task